A water gap as a diode

Aluminium valve cells and polarity spark gaps as rectifiers — CLAUDE.md §3g, §3h

kronspice simulation report · 2026-11-07 · declared cell and gap parameters

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Summary

Can a water gap act like a diode because of its two electrode materials? Yes, and it was done over a century ago. A **valve metal** (aluminium) in water grows an insulating oxide when it is the positive electrode and conducts freely when it is the negative one. Paired with an ordinary metal (lead, iron) it is the aluminium electrolytic rectifier of Pollak (1896) and Nodon (1899): about 140 V per cell, about 75–80 % efficient (*Encyclopaedia Britannica* 1911, “Transformers”). At kilovolts the water itself breaks down in a spark; there the one-way action comes from the **shape** of the electrodes (a sharp positive point breaks down at a lower voltage than a sharp negative one), the polarity spark gap. The example models both, builds rectifiers from them, and compares them in a pulse test and a frequency sweep.

- **Aluminium against any ordinary metal rectifies:** forward/reverse 1.37×10^{4} at 100 V for lead, iron or silver-plated iron against aluminium; two ordinary metals are symmetric; aluminium against aluminium blocks both ways.
- **Best rectifier:** a bridge of four aluminium–lead cells at 100 V peak gives 88.0 V with 5.21 V ripple at 85.0 % efficiency, at the top of the 1911 account's 75–80 %.
- **The oxide film limits frequency.** It is a 4.38 µF capacitor; the bridge falls to 17.2 % efficiency at 20 kHz (84.1 % at 1 kHz).
- **Spark-gap bridges need a polarity ratio above 2.** With the declared ratio 2 the bridge never strikes (4 kV) or breaks down backwards (6 kV); with ratio 3 it rectifies at 67.9 %. Both breakdown voltages are extrapolated and the ratio is declared.
- **Fastest amps:** the same 10 J gives 4247 A in 0.108 µs through a 3 mm water gap, against 470 A in 9.22 µs through a lead–aluminium cell, but the cell hands over all the energy and the gap leaves 8.10 J in the capacitor.
- **Spark gaps work only at mains frequency:** the water-gap bridge gives 0.247 A/V at 50 Hz and 0 from 300 Hz up; the cells hold about 0.459 A/V to 100 kHz, but their efficiency falls to 34 %.

Model

- **Water cell** (*cell.go*): plates 100 cm², 1 cm apart, water 0.1 Ω (phosphate solution), decomposition onset 2.0 V; aluminium's oxide blocks the reverse direction and sparks over at 140 V; the film is a 4.38 µF capacitor. Each I–V curve is checked against a kronspice DC sweep of the same element.
- **Rectifiers** (*rectifier.go*): kronspice .TRAN with the ledger on; load 50 Ω with a 2,200 µF filter, source resistance 0.5 Ω.
- **Polarity spark gap** (*gap.go*, *gapsim.go*): 3 mm point–plane gap in pure water, breakdown +2415 V with the point positive (extrapolated from Stygar et al. 2006, sub-microsecond data, to 60 Hz) and −4830 V with it negative (ratio 2, declared from J. C. Martin's constants, to be checked); leak 57.9 MΩ; arc 50 V + 0.5 Ω. A spark gap is a switch with memory, so it is stepped by a small nodal solver of its own, which matches kronspice below breakdown (TestGapSolverTwin).
- **Recovery:** declared as $1 - e^{-t/1\ \mathrm{ms}}$; a hot gap never re-strikes below the Paschen minimum of water vapour, 480 V (Škoro et al. 2011).

Part 1 — electrode pairs in water

Current from the first electrode to the second.

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pair	+5 V	−5 V	+100 V	−100 V	−200 V	fwd/rev at 100 V	twin
lead | water | aluminium	30 A	−0.00357 A	980 A	−0.0714 A	−28.7 A	1.37×10^{4}	6.8e-17
iron | water | aluminium	30 A	−0.00357 A	980 A	−0.0714 A	−28.7 A	1.37×10^{4}	6.8e-17
copper with silver tips | water | aluminium	3 A	−0.00357 A	98 A	−0.0714 A	−20.1 A	1.37×10^{3}	3.1e-16
silver-plated iron | water | aluminium	30 A	−0.00357 A	980 A	−0.0714 A	−28.7 A	1.37×10^{4}	6.8e-17
iron | water | copper	30 A	−30 A	980 A	−980 A	-1.98×10^{3} A	—	0.0
lead | water | iron	30 A	−30 A	980 A	−980 A	-1.98×10^{3} A	—	0.0
aluminium | water | aluminium	0.00357 A	−0.00357 A	0.0714 A	−0.0714 A	−28.7 A	—	4.7e-15

A valve metal against an ordinary metal rectifies; any ordinary metal, silver included, works as the partner, because the one-way action comes from the aluminium's oxide. Two ordinary metals do not rectify; they make a weak battery (iron | water | copper 0.789 V open circuit; lead | water | iron −0.321 V). Two valve metals block both ways. Past the oxide's sparking voltage (140 V) the blocked direction sparks over, like a Zener diode.

Part 2 — aluminium–lead water cells as rectifiers

circuit	Vdc	ripple	load	efficiency	reverse peak	ledger
half-wave, 1 cell, 100 V peak, 60 Hz	76.2 V	27.28 V	118.3 W	20.5 %	167 V (sparks over)	4.6e-09
half-wave, 2 cells in series, 100 V peak	83.3 V	10.91 V	139.0 W	84.0 %	183 V	−1.6e-09
bridge, 1 cell per arm, 100 V peak, 60 Hz	**88.0 V**	**5.21 V**	**154.8 W**	**85.0 %**	**93 V**	**1.1e-08**
bridge, 1 cell per arm, 170 V peak (120 V rms)	145.1 V	9.69 V	421.1 W	40.1 %	153 V (sparks over)	1.2e-07
bridge, 2 cells per arm, 170 V peak	146.5 V	8.36 V	429.4 W	85.6 %	156 V	3.5e-09
bridge, 1 cell per arm, 100 V peak, 1 kHz	88.3 V	0.32 V	155.9 W	84.1 %	91 V	2.5e-08
bridge, 1 cell per arm, 100 V peak, 20 kHz	85.9 V	0.02 V	147.5 W	17.2 %	89 V	2.9e-09
bridge, copper with silver tips | aluminium	79.2 V	4.04 V	125.6 W	77.5 %	88 V	7.9e-09
bridge, silver-plated iron | aluminium	88.0 V	5.21 V	154.8 W	85.0 %	93 V	1.1e-08

Efficiency = load power ÷ source power over the last cycles. 60 Hz unless stated.

- A filtered half-wave rectifier sees twice the peak in reverse, so one cell sparks over; two in series hold, as in the historical practice of “two or more valves in series for higher pressures”.
- At 20 kHz the oxide film's capacitance passes current both ways and the efficiency collapses, the 1911 article's warning that these rectifiers are unsuitable for high frequencies.
- 85 % sits at the top of the 75–80 % the 1911 account gives; the cell's leakage and resistance are declared.

Silver against aluminium

Both silver versions rectify, but when a cell conducts the silver is the positive electrode, and a positive metal electrode in water is eaten away. If all the current dissolved silver (Faraday's law, the worst case):

silver electrode (declared size)	silver	charge it passes	lasts at the bridge's current
copper with 1 mm solid silver tips on 10 % of the plate	10.5 g	9.38×10^{3} C	3.3 h (0.79 A per cell)
iron with 10 µm silver plating	1.05 g	938 C	18 min (0.88 A per cell)

After that the copper or iron underneath carries the current and corrodes. This is why Pollak used lead: as the positive electrode it forms a stable lead-dioxide skin instead of dissolving.

Part 3 — polarity spark gaps in water as rectifiers

A one-way spark needs a forward voltage above +Vb and a reverse voltage below −Vb·ratio. With a filter capacitor a half-wave rectifier sees 2·Vpeak in reverse, and a bridge needs 2·Vb forward (two gaps in series) while each blocked gap sees the full output: both need a polarity ratio above 2.

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circuit	ratio	Vdc	ripple	efficiency	reverse peak	strikes f/b	result
half-wave, resistive load, 4 kV peak	2.0	900 V	4014 V	96.1 %	3932 V	8617/0	rectifies
half-wave, filter capacitor, 4 kV peak	2.0	366 V	6014 V	7.0 %	4830 V	21/20	breaks down backwards
bridge, filter capacitor, 4 kV peak	2.0	0 V	0 V	0.0 %	2029 V	0/0	never strikes
bridge, filter capacitor, 6 kV peak	2.0	1837 V	7173 V	0.1 %	9474 V	540/520	breaks down backwards
bridge, filter capacitor, 6 kV peak	**3.0**	**3107 V**	**4519 V**	**67.9 %**	**5965 V**	**6/0**	**rectifies**

A spark gap is not a low-drop diode: it conducts only once the voltage passes its breakdown (2.4 kV here), so the output sags until the source exceeds the load voltage by the gaps' breakdown, then jumps back up. **A bench measurement** of both breakdown voltages and their ratio decides whether a water spark-gap bridge can work.

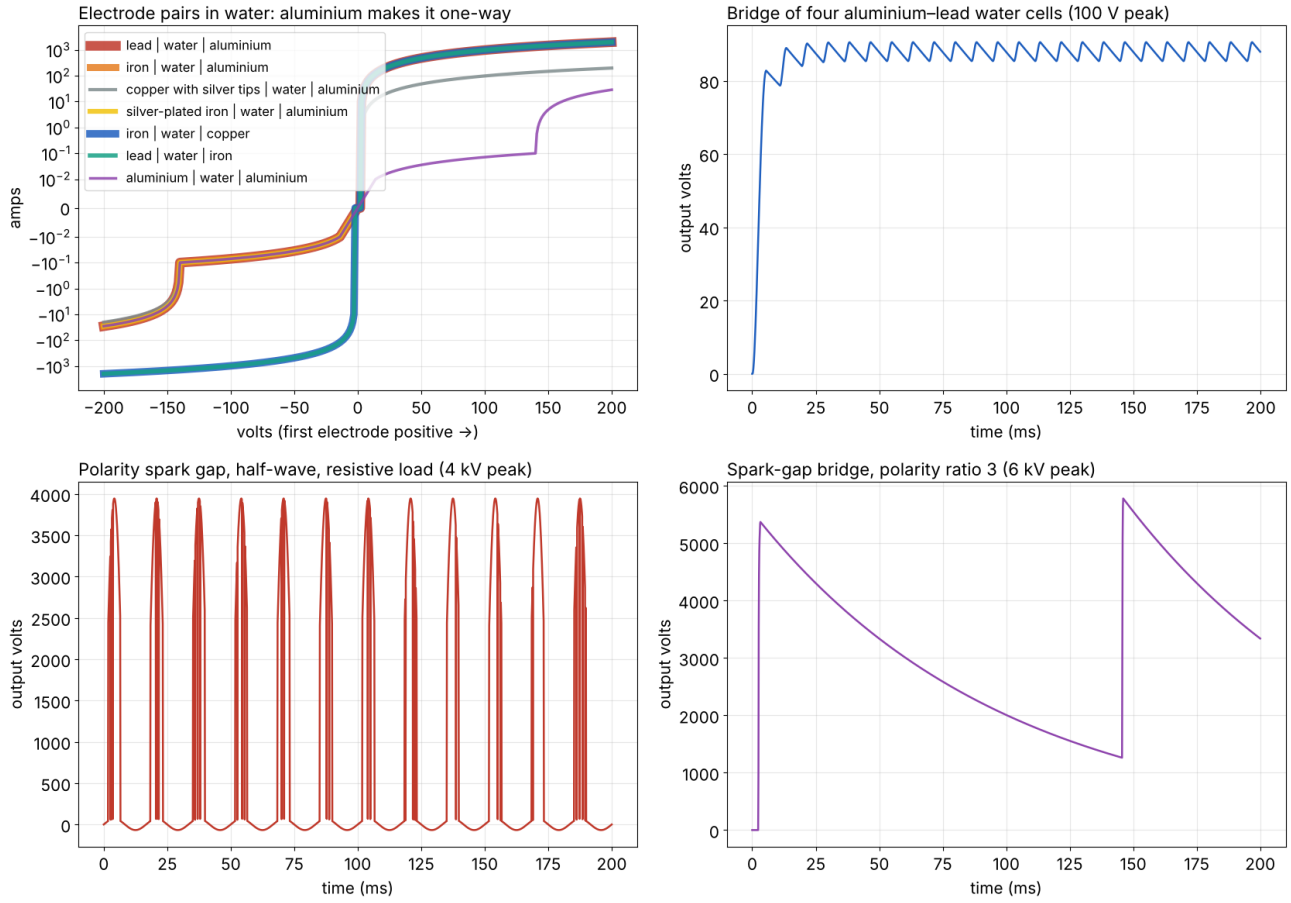


Figure 1. Top left: I–V curves of the seven electrode pairs (symmetric-log current axis); aluminium against an ordinary metal conducts forward and blocks to 140 V in reverse, aluminium against aluminium blocks both ways. Top right: output of the bridge of four aluminium–lead cells at 100 V peak. Bottom left: polarity spark gap, half-wave, resistive load, 4 kV peak (pulsed output). Bottom right: spark-gap bridge with polarity ratio 3 at 6 kV peak: the output decays until the gaps strike again.

Part 4 — pulse test: which gives the fastest amps?

The same 10 J from a charged capacitor through each device, a 1 µH loop and a 0.1 Ω load (declared).

device	charge	C	peak	rise (10–90 %)	to load	in device	left in C
lead | water | aluminium	100 V	0.002 F	470 A	9.22 µs	4.80 J	5.19 J	0.00 J
silver-plated iron | water | aluminium	100 V	0.002 F	470 A	9.22 µs	4.80 J	5.19 J	0.00 J
copper with silver tips | water | aluminium	100 V	0.002 F	195 A	2.22 µs	0.86 J	8.82 J	0.32 J
water polarity gap, 3 mm	**41000 V**	**1.19×10^{-8} F**	**4247 A**	**0.108 µs**	**0.31 J**	**1.59 J**	**8.10 J**

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device	charge	C	peak	rise (10–90 %)	to load	in device	left in C
mist-filled air gap, 1 mm	8000 V	3.13×10^{-7} F	3510 A	0.505 µs	1.09 J	5.55 J	3.36 J

Spark gaps give the highest and fastest current because they run at kilovolts. The cells give hundreds of amps in microseconds, limited by the water's resistance, and hand over all the energy. A gap's arc goes out at the first current zero, so energy still ringing in the capacitor stays there. The water gap's charge follows Stygar's relation at 1 µs, inside its measured range; the mist gap uses the fog factor from *water_lamp* (extrapolated).

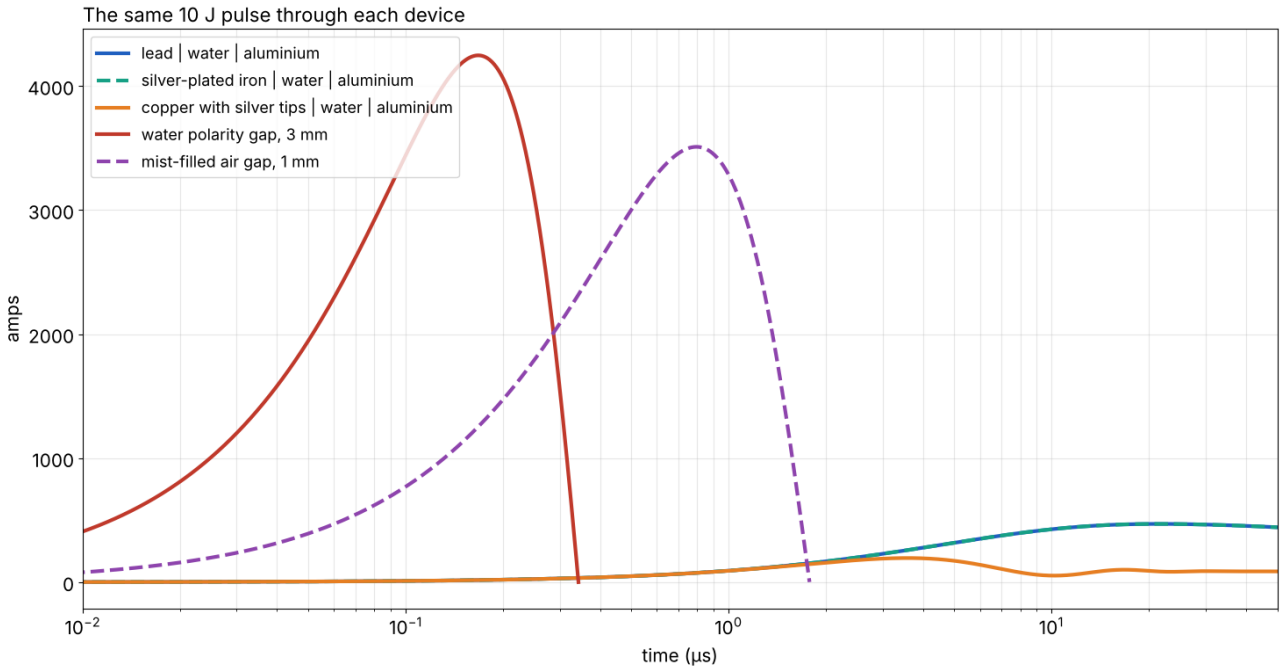


Figure 2. Current through each device for the same 10 J pulse (log time axis, µs). The water gap peaks at about 4.2 kA and the mist gap at 3.5 kA within a microsecond; the cells rise to hundreds of amps over about 10 µs.

Part 5 — frequency sweep: amps per volt

Every device in the same bridge (source 0.1 Ω, load 1 Ω, no filter), 50 Hz to 100 kHz. Amps per volt = mean load current ÷ source peak voltage. A perfect diode bridge gives 0.579 A/V at every frequency.

device	50 Hz	100	300	1k	3k	10k	30k	100k
lead | aluminium (100 V), A/V	0.459	0.459	0.459	0.459	0.460	0.463	0.481	0.508
efficiency	73 %	73 %	73 %	73 %	73 %	71 %	60 %	34 %
silver-plated iron | aluminium (100 V)	same as lead							
copper with silver tips | aluminium (100 V), A/V	0.192	0.192	0.192	0.192	0.193	0.193	0.195	0.191
efficiency	31 %	31 %	31 %	30 %	30 %	29 %	22 %	8 %
water gap, ratio 3, 6 kV fixed, A/V	0.247	0.198	0	0	0	0	0	0
efficiency	47 %	47 %	0 %	0 %	0 %	0 %	0 %	0 %
water gap, ratio 3, voltage raised with f, A/V	0.239	0.240	0	0	0	0	0	0
efficiency	47 %	47 %	0 %	0 %	0 %	0 %	0 %	0 %

- **Water cells** hold steady to about 3 kHz. Above that the oxide film passes current both ways: the mean current holds up but the efficiency collapses (34 % at 100 kHz).
- **Water spark gaps** work only at mains frequencies. At a fixed 6 kV the breakdown voltage rises with frequency ($E{\cdot}\tau^{1/3}$, extrapolated) until, above 100 Hz, the gaps never strike. With the voltage raised to keep them striking, above about 100 Hz the gaps that just carried an arc have not recovered and break down backwards: the bridge shorts. Spark gaps are pulse switches, not high-frequency rectifiers.

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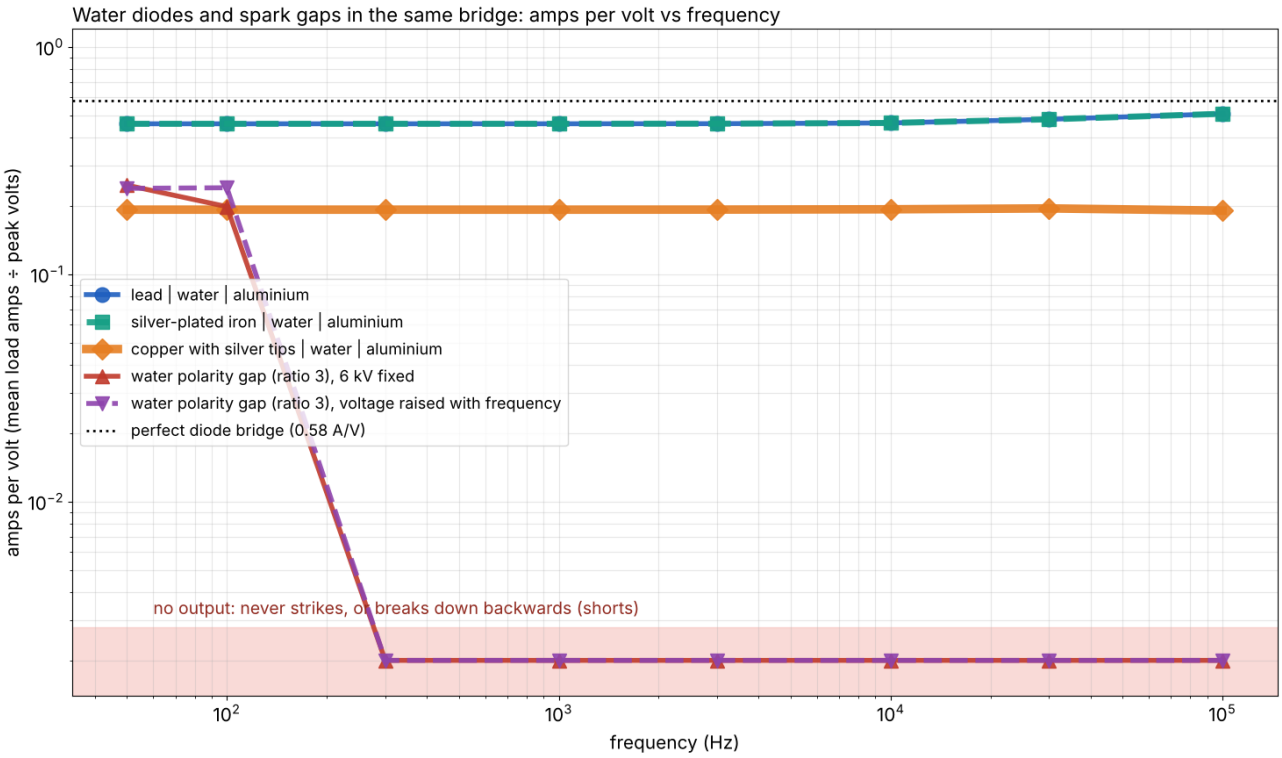


Figure 3. Amps per volt against frequency for every device in the same bridge. The cells stay flat (lead and silver-plated iron overlap); the water gaps drop to no output above 100 Hz (shaded band). Dotted line: perfect diode bridge, 0.58 A/V.

Ledger and validation

- Every rectifier run's energy ledger closes to about 10^{-7} or better (largest listed 1.2×10^{-7}).
- Each cell I–V curve matches a kronspice DC sweep of the same element (twin $\le 4.7\times10^{-15}$).
- The spark-gap solver matches kronspice below breakdown (TestGapSolverTwin).
- The water and the metals add no energy: every watt at the load comes from the AC source; the rest is heat and, for the cells, some electrolysis.

Limits of the model

- Cell leakage and resistance are declared values; oxide sparking at 140 V follows the 1911 figure per cell.
- The water gap's positive breakdown is extrapolated from sub-microsecond data to 60 Hz; the polarity ratio of 2 is declared (to be checked); the mist gap's fog factor is extrapolated.
- Gap recovery (1 ms) is declared.
- Silver lifetimes are a worst case (all current dissolving silver); part of it may evolve oxygen or form a film instead.

Reproducing the results

```
go run ./examples/sparkgap_diode -viz examples/sparkgap_diode/viz   # a few seconds
python3 examples/sparkgap_diode/viz/render.py
go test ./examples/sparkgap_diode
```

Bridges that combine these cells and gaps for particular jobs (surge protection, a spark pulser, a voltage doubler, the oxide film as a varactor, a frequency doubler) are in *examples/sparkgap_diodebridge*. Sources: Encyclopaedia Britannica 11th ed. (1911) vol. 27 “Transformers”; W. A. Stygar et al., Phys. Rev. ST Accel. Beams 9, 070401 (2006); M. Škoro et al., Phys. Rev. E 84, 055401 (2011); L. Liu et al. (2018); CRC Handbook electrochemical series.

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